Aditya Kandukoori

adirocknbolt@gmail.com • (240)-490-0448

EDUCATION

Poolesville High School
September 2024 - May 2028
Science, Math, Computer Science Program
Relevant Coursework: Algorithm Data (Magnet), Fundamentals of Computer Science (Magnet), Foundations of Technology, Magnet Analysis (Calculus), Adv Magnet Physics, Adv Magnet Chemistry, Adv Biology, Adv Earth Space and Systems,

EXPERIENCE

MoCo Innovation (STEM Nonprofit Organization)
President
August 2025 – Present

- Lead a STEM nonprofit providing technology education to underrepresented students.
- Raised **$32,000+** in sponsorships to expand programming across multiple schools.
- Taught coding and computational thinking to elementary and K–12 students through in-person and online programs.

PROJECTS

Comparative Analysis of Digital Tools and Traditional Teaching Methods in Educational Effectiveness
- Conducted a study comparing digital math tools and traditional instruction, finding significantly higher learning gains with technology-based methods**. Link.**
Biasmap - Analyzing societal bias in large language models by comparing model outputs with real-world sentiment from news and online discourse**.**
Indirect Measurement - Designed a triangulation-based device combining a clinometer and level to measure angles, distances, and object heights.
Catapult - Designed and built a catapult, a modular catapult meeting strict constraints, demonstrating skills in engineering, teamwork, and problem-solving.
Falconwing - Built a 3D-printed Arduino-based prosthetic arm using MyoWare EMG sensors to control a grasping mechanism via muscle signals along with my team.
Strawberry DNA Extraction Experiment - Designed and carried out experiments to extract DNA from strawberry samples, using techniques such as tissue pulverization, micropipetting, and centrifugation to isolate genetic material. Carefully documented procedures, observations, and conclusions in a laboratory journal.
Unknown Bacterial Culture Identification Project - Isolated and identified unknown bacteria from a mixed culture using electrophoresis and documented results in a research report.
UVC Exposure and Sunscreen Efficacy Experiment - Conducted an experimental study examining how SPF 30 sunscreen affects bacterial survival under UVC exposure using Rhodospirillum rubrum. Analyzed lethal exposure thresholds and documented methods, results, and conclusions in a detailed research report

SKILLS

- Python, Java, Microsoft Excel, data analysis, Arduino & embedded systems, sensor integration, experimental biology, electrophoresis, engineering prototyping, AI model analysis, technical writing, teamwork

HONORS AND SERVICE

- **Research publication on arXiv**
- **200+ volunteer hours (food drives, summer school volunteering)**